

Analogous Dynamics of Couette–Poiseuille Flows and Adverse-Pressure-Gradient Turbulent Boundary Layers

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Abstract

Turbulent boundary layers (TBLs) subjected to adverse pressure gradients (APGs) are ubiquitous in engineering and geophysical flows, yet they remain challenging to analyze and model due to their pronounced spatial development and sensitivity to upstream conditions. In contrast, turbulent Couette–Poiseuille (C-P) flow, a statistically one-dimensional flow between two parallel walls driven by both wall motion and pressure gradient, offers a simplified framework for studying wall-bounded turbulence subjected to APGs.

In this study, we compare equilibrium APG TBLs with constant Clauser parameters to equilibrium turbulent C-P flows driven by equivalent pressure gradients. Notable similarities between turbulent C-P flows and APG TBLs are observed, including profiles of turbulence statistics and characteristics of flow structure. By drawing analogies and constructing similarity transformations between the two systems, we demonstrate that turbulent C-P flows can serve as a simplified model for understanding and modeling the complex dynamics of APG TBLs.

This work was supported by National Science Foundation (NSF) grant No. 2152705. Computer time was provided by the Discover project at Pittsburgh Supercomputing Center through allocation PHY240020 from the Advanced Cyberinfrastructure Coordination Ecosystem: Services & Support

(ACCESS) program, which is supported by NSF grants No. 2138259, No. 2138286, No. 2138307, No. 2137603, and No. 2138296. Z. Shu acknowledges Professor Chunxiao Xu and the Tsien Excellence in Engineering Program for providing the opportunity and financial support for academic exchange.